

Bequests, Fertility and Barriers to Entrepreneurship:

The Intergenerational Transmission of Economic Inequality

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Abstract

This paper studies how parental fertility and bequest decisions jointly shape children's entrepreneurial opportunities under financial frictions. While existing work emphasizes bequests, it largely abstracts from fertility, which determines how parental resources are divided among children. I develop an overlapping-generations model with endogenous fertility, bequests, and entrepreneurship choice, calibrated to U.S. data. The model shows that fertility-driven dilution of intergenerational transfers interacts with borrowing constraints to distort entrepreneurial choices and reduce efficiency. Eliminating financial frictions in the US increases income per capita by 13 percent, but slightly lowers the entrepreneurship rate, increases inequality, and reduces intergenerational mobility.

Keywords: Entrepreneurship, Financial Frictions, Bequest, Fertility

JEL Classification: O16, E24

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1 Introduction

Entrepreneurship is crucial for economic development, but many entrepreneurs face borrowing constraints that limit their ability to start or grow businesses (Buera, 2008; Evans and Jovanovic, 1989). Intergenerational transfers can help alleviate financial stress, and existing work has therefore emphasized their importance as determinants of entrepreneurial opportunity. However, current literature has largely abstracted from parental fertility choices, which determine resources per child. Empirical evidence has shown that wealthier households tend to have fewer children (Becker, 1960; Borg, 1989; Docquier, 2004; Jones et al., 2008) and can concentrate more resources and leave larger bequests on each child. Individuals with high-income parents typically encounter fewer financial barriers and are more likely to become entrepreneurs. Ignoring the endogenous fertility channel would bias the estimated influence of financial frictions on economic growth and inequality.

This paper examines how parents' decisions regarding fertility and bequests impact their children's ability to become entrepreneurs when financial markets are imperfect. Fertility determines how parental resources are diluted across children, and financial frictions make inherited resources matter for entrepreneurial entry and firm scale. Parental bequests and fertility choices interact with financial frictions to distort children's entrepreneurship decisions. Highly talented individuals from larger, less wealthy families may be unable to obtain enough capital to operate optimally, whereas less talented individuals from smaller, wealthier families may be able to enter and expand. This mechanism generates a mismatch between entrepreneurial talent and capital, reducing aggregate efficiency. Because entrepreneurship is a key channel for wealth accumulation and many of the wealthiest individuals are business owners (Cagetti and De Nardi, 2006), these distortions can also amplify income inequality and reduce intergenerational mobility.

To study these dynamics, I develop an overlapping generations model where parents

choose how many children to have and how much wealth to transfer to all the children. There is no heterogeneity within the family. Children inherit the bequest from their parents as the assets and partially inherit the entrepreneurial talent following an AR(1) process. In a benchmark economy without financial frictions, individuals can become entrepreneurs as long as their entrepreneurial talent is high enough, and all entrepreneurs operate at their optimal scale. I then introduce financial frictions arising from limited contract enforceability, following Buera et al. (2011), which generates endogenous borrowing constraints that depend on individuals' wealth and entrepreneurial talent. In this environment, parental fertility and bequest choices directly shape children's entrepreneurial opportunities by determining their access to startup capital.

After solving the model, I calibrate it using data from the Panel Study of Income Dynamics (PSID). The calibration proceeds in two stages. In the first stage, I estimate individual-level income and consumption profiles over the life cycle using fixed-effects regressions, which allows me to construct expected lifetime income and consumption. I also use asset holdings at age 25, when individuals are unlikely to have accumulated substantial wealth on their own, as a proxy for intergenerational transfers. Macro-level parameters are then calibrated by Generalized Method of Moments (GMM) to match key aggregate moments, including the entrepreneurship rate and the income distribution. In the second stage, I use the estimated income profiles from the first stage to calibrate utility parameters by matching the mean of log consumption and the distribution of fertility.

The calibration suggests that the U.S. credit market is relatively well functioning, in line with the baseline assumptions in the cross-country studies relating to financial development (Buera et al., 2011). The model also helps reconcile the evidence in Hurst and Lusardi (2004), who find that entrepreneurial entry is related to wealth only near the top of the wealth distribution. In my framework, this pattern arises because borrowing constraints depend on the interaction between entrepreneurial talent and wealth. The borrowing constraints bind

mainly for high-talent individuals with low wealth and for low-talent individuals with high wealth who can leverage their assets into entry.

Counterfactual analysis shows that reducing financial frictions, relative to the baseline model calibrated to the US, improves economic development, but at the cost of slightly reducing entrepreneurship and increasing income inequality. This occurs because more developed financial markets enable highly productive individuals to attract more capital by offering higher interest rates, thereby amplifying returns on wealth and crowding out marginal entrepreneurs. These findings underscore the importance of designing finance, tax, and transfer policies that account for both economic development and concerns about income concentration.

Another counterfactual exercise shows that shutting down the endogenous fertility channel, as is common in the existing literature, can lead to a misleading estimation of inequality and intergenerational mobility. When fertility is treated as exogenous, the model fails to capture fertility's dilution effect on family resources and its influence on wealth distribution. This result highlights the importance of considering the endogenous fertility channel in policy evaluations.

This paper contributes to three strands of literature. First, it contributes to the literature on fertility and intergenerational resource allocation by emphasizing fertility's dilution effect on intergenerational monetary transfers and, thereby, its impact on children's entrepreneurial opportunities. Second, it contributes to the literature on entrepreneurship and financial frictions by showing that family size is an important determinant of the initial resources available for entrepreneurs to overcome borrowing constraints. Third, it contributes to the literature on inequality and intergenerational mobility by highlighting how fertility-driven dilution of family resources interacts with the leverage effect of entrepreneurship to shape the persistence of income inequality across generations when financial frictions exist.

The remainder of the paper is structured as follows. Section 2 reviews the related literature and outlines the paper’s contribution. Section 3 presents the frictionless benchmark model. Section 4 introduces financial frictions into the framework. Section 5 describes the data and calibration strategy. Section 6 reports the counterfactual analysis. Section 7 concludes.

2 Related Literature

The key mechanism in this paper relates to the literature on fertility-driven dilution of family resources. The classical quantity–quality trade-off literature defines “quality” primarily in terms of children’s human capital investment (Becker, 1960; Docquier, 2004) and fertility choices often affect intergenerational mobility and inequality through human capital accumulation and its impact on labor income (Ehrlich and Kim, 2007; Moav, 2005). A related strand of literature emphasizes that fertility also dilutes wealth transfers among children and thereby shapes intergenerational mobility and income inequality through wealth accumulation (Aso, 2024; Cooke et al., 2017; Pestieau, 1984). Complementary work studies how monetary transfers affect wealth and income distributions without explicitly modeling fertility (De Nardi, 2004; De Nardi and Yang, 2016). More recently, scholars have combined fertility choices with both human capital investment and monetary transfers to study how family choices influence social mobility in a more comprehensive way (Daruich and Kozlowski, 2016), though abstracting from the entrepreneurship channel.

However, the influence of intergenerational wealth transmission on mobility and inequality is not limited to the direct effect of inherited wealth, as wealth can have a leverage effect through entrepreneurship. When individuals receive monetary transfers from parents, they can relax borrowing constraints and finance a larger amount of capital when starting a business. Empirically, the relationship between wealth and entrepreneurial entry is nu-

anced. Though Hurst and Lusardi (2004) argues that wealth holdings only influence the entrepreneurial entry at the top of the wealth distribution, Fairlie and Krashinsky (2012) shows that the irrelevance between business entry and wealth is due to the changing proportion of job losers at different asset levels. After examining the subsamples of job losers and non-job losers separately, they find an increasing rate of entrepreneurial entry along the asset distribution. Taken together, this evidence supports the view that wealth and financial frictions meaningfully shape entrepreneurship.

Entrepreneurship, in turn, is a key driver of wealth concentration and persistence at the top of the income and wealth distributions (Brüggemann, 2021; Quadrini, 1999, 2000; Scheuer, 2014; Terajima, 2006). As a result, the monetary transfers within the family across generations lead to the intergenerational transmission of inequality: individuals from rich families are more likely to overcome entrepreneurial financial barriers and become entrepreneurs, thereby remaining at the top of income and wealth distributions through business ownership.

Existing quantitative work captures the influence of bequest on entrepreneurship and, consequently, income and wealth distributions, but typically abstracts from the endogenous fertility channel. Cagetti and De Nardi (2006) models intergenerational transmission through discount factors but does not explicitly include endogenous bequests. Ivanova (2023) examines how family monetary transfers alleviate financial barriers to entrepreneurship and shape wealth and income distributions in a dynamic model, but it omits the dilution effect of fertility on family resources. However, empirical studies show that family structure and size can affect entrepreneurship (Vladasel, 2023), aligning with the dilution mechanism in family resource allocation.

The main contribution of this paper is to integrate these strands of research by studying how fertility-driven dilution of family monetary transfers interacts with credit constraints to shape entrepreneurship choices and how these dynamics map into intergenerational mobility

and the distributions of income and wealth through entrepreneurship's leverage effect.

3 Frictionless Model

I first build a model without financial frictions as the benchmark. I adopt a discrete-time OLG model where people experience two stages of life: child and adult. The adults make entrepreneurship choices and maximize their utility. The children consume passively.

3.1 Utility Maximization

After knowing their income, parents choose consumption c_{it} , total bequest to all children m_{it} and the number of children n_{it} to maximize their utility:

$$\max_{\{c_{it}, n_{it}, m_{it}\}} \left(\theta_1 c_{it}^{1-\frac{1}{\epsilon}} + \theta_2 m_{it}^{1-\frac{1}{\epsilon}} + (1 - \theta_1 - \theta_2) n_{it}^{1-\frac{1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \quad (1)$$

$$\text{Subject to: } c_{it} + m_{it} = (1 - \tau n_{it}) y_{it}$$

Parameter ϵ is the elasticity of substitution between consumption, total bequest, and fertility. θ_1 and θ_2 represent how much people value consumption and bequest respectively. y_{it} is individual i_t 's income. τ is the time cost of having one child. To raise n_{it} children, the parent needs to pay $\tau \times n_{it}$ fraction of their income y_{it} .

The solution to the utility maximization problem is listed as follows:

$$c_{it}^* = \frac{1}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1 - \theta_1 - \theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \quad (2)$$

$$m_{it}^* = \frac{(\frac{\theta_2}{\theta_1})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1 - \theta_1 - \theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \quad (3)$$

$$n_{it}^* = \frac{(\frac{1 - \theta_1 - \theta_2}{\theta_1 \tau})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1 - \theta_1 - \theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it}^{1-\epsilon} \quad (4)$$

Consumption and bequest are increasing in an individual's income. The monotonicity of the number of children depends on the value of ϵ . When $\epsilon \leq 1$, n_{it} increase in y_{it} . However, if $\epsilon > 1$, n_{it} will first increase in income when $y_{it} < (\frac{1 - \theta_1 - \theta_2}{\theta_1 \tau}) (\frac{1 + (\frac{\theta_2}{\theta_1})^\epsilon}{\epsilon - 1})^{\frac{1}{\epsilon}}$. As income increases above the threshold, the number of children will decrease in income due to the increased child cost. Figure 1 presents the relationship between income and the number of children for different values of ϵ .

I define $b_{it} = \frac{m_{it}}{n_{it}}$ as the average bequest, and each child within the family obtains the same amount of bequest b_{it} . When the child grows up and becomes an adult, they will inherit the amount of b_{it} as their asset. In aggregate, for the next generation $t + 1$, a fraction of $\frac{n_{it}}{\sum_{it}(n_{it})}$ people will receive b_{it} amount of bequests as their assets. Parents' optimal bequest and fertility choices determine the distribution of assets for the next generation.

3.2 Individuals' Setup

When the children grow up, they inherit bequests from their parents as assets:

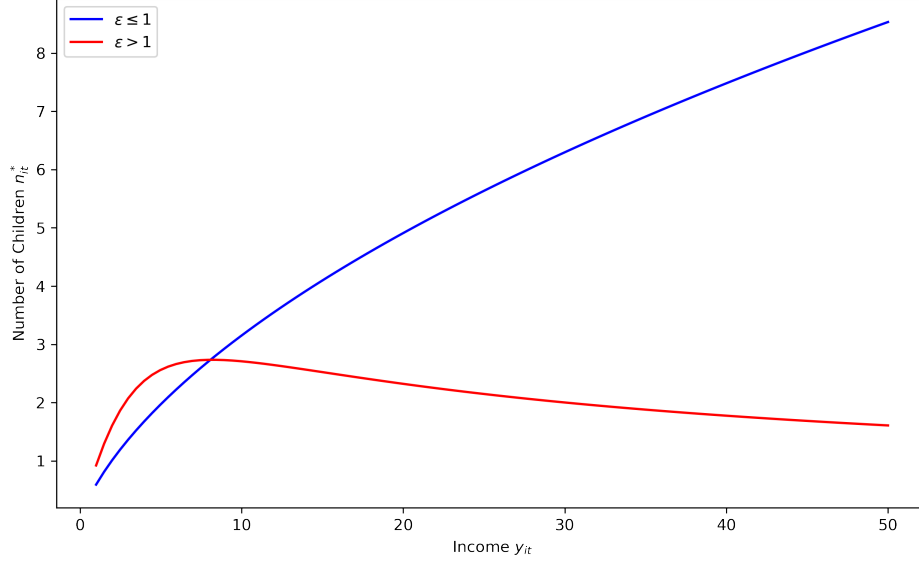


Figure 1: Relationship between Income and Number of Children

$$a_{\text{children},t} = b_{\text{parent},t-1} \quad (5)$$

At the same time, entrepreneurial ability is given by the following equation:

$$\log(z_{\text{children},t}) = \rho \log(z_{\text{parent},t-1}) + \bar{z} + \epsilon_{it} \quad (6)$$

$\epsilon_{it} \sim N(0, 1)$. Each individual's talent is partially inherited from the parent and $0 < \rho < 1$. In the long run, the individual-level talent would converge to $e^{\frac{\bar{z}}{1-\rho}}$.

3.3 Occupation

After knowing the asset level a_{it} and entrepreneurial talent z_{it} , the individual i_t first faces the following firm production problem:

$$\max_{\{K_{it}, L_{it}\}} \pi_{it} = (A \times z_{it})^{1-\alpha-\beta} (K_{it})^\alpha L_{it}^\beta - R_t K_{it} - W_t L_{it} \quad (7)$$

After solving the previous problem, the entrepreneurs know their optimal capital level, labor level, and the profit:

$$K_{it}^* = \left(\frac{\alpha}{R_t}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} A z_{it} \quad (8)$$

$$L_{it}^* = \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{1-\alpha}{1-\alpha-\beta}} A z_{it} \quad (9)$$

$$\pi_{it}^* = A z_{it} \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} (1 - \alpha - \beta) \quad (10)$$

The depreciation rate of capital is assumed to be 1.

As an entrepreneur, the individual can also earn interest on the asset from the bank. Consequently, the total income of an entrepreneur should be:

$$y_{it}^E = \pi_{it}^* + R_t a_{it} \quad (11)$$

On the other hand, an individual provides one unit of labor inelastically as a worker. Workers are matched with the entrepreneurs randomly, and each worker earns the average expected wage in the labor market. The income of a worker is:

$$y_{it}^W = W_t + R_t a_{it} \quad (12)$$

By comparing y_{it}^E and y_{it}^W , the individual makes an occupation choice which is summarized in equation (13). An individual chooses to become an entrepreneur if and only if $y_{it}^E \geq y_{it}^W$.

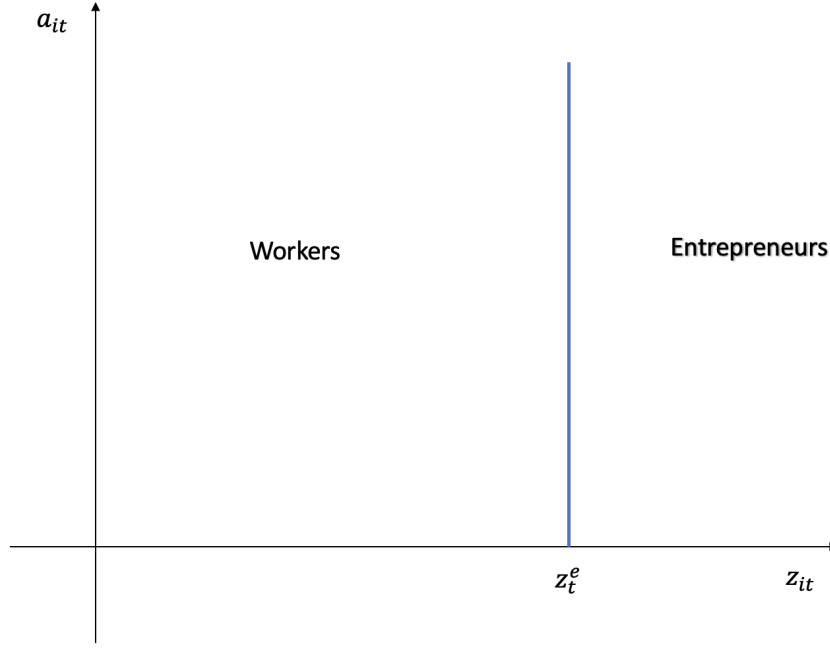


Figure 2: Occupation Choices in the Frictionless Model

After solving the occupation problem, I get the entrepreneurial cutoff z_t^e in equation (14).

$$o_{it} = \begin{cases} 1 & \text{if } y_{it}^E \geq y_{it}^W \\ 0 & \text{if } y_{it}^E < y_{it}^W \end{cases} \quad (13)$$

$$z_t^e = \frac{1}{A} \frac{\beta}{1 - \alpha - \beta} \left(\frac{R_t}{\alpha} \right)^{\frac{\alpha}{1 - \alpha - \beta}} \left(\frac{W_t}{\beta} \right)^{\frac{1 - \alpha}{1 - \alpha - \beta}} \quad (14)$$

I ignore the subscript i here because the cutoff doesn't depend on individual-level characteristics. Individuals with entrepreneurial talent higher than z_{it}^e choose to become entrepreneurs with income y_{it}^E . Those beneath the threshold will become a worker with income y_{it}^W . Picture 2 shows individuals' occupation choices in a frictionless model.

3.4 Equilibrium

Let $\mathbf{x} = (a, z)$ be the state variable in the economy. Following the solutions to the utility maximization problem, the decision rules for occupation and income, and the AR(1) process for entrepreneurial talent, I can calculate the transition function $p(\mathbf{x}'|\mathbf{x})$, which governs the conditional probability of the next state based on the current state.

The stationary equilibrium is defined by a stable distribution $d^*(\mathbf{x})$ of individuals over state variables $\mathbf{x}$; optimal decision rules $c(\mathbf{x})$, $n(\mathbf{x})$ and $m(\mathbf{x})$; the occupational choice $o(\mathbf{x})$; entrepreneurs' optimal solutions $K(\mathbf{x})$ and $L(\mathbf{x})$; interest rate R , wage W . In equilibrium, the following conditions must be satisfied:

- (a) Adults maximize utility subject to budget constraints.
- (b) Entrepreneurs maximize profits.
- (c) Individual makes the optimal occupational choices based on (13).
- (d) Labor and capital markets clear. The labor supplied by all workers should be equal to all labor employed by entrepreneurs. The total assets equal all capital employed by entrepreneurs:

$$\int_{i_t \in W} di_t = \int_{i_t \in E} L_{it}^* di_t \quad (15)$$

$$\int_{i_t} a_{it} di_t = \int_{i_t \in E} K_{it}^* di_t \quad (16)$$

- (e) d^* is stable.

4 Model with Financial Frictions

The structure of the model with financial frictions is similar to the benchmark model. Parents solve the utility maximization problem in (1) by choosing the optimal consumption, bequest, and fertility choices. When children grow up, they inherit the bequest from their parents as the asset, following equation (5), and get entrepreneurial talent, as specified in equation (6). The only difference is that now entrepreneurs can either default on the contract or pay back the interest to the bank. The bank will set a limit on the capital it will lend to each individual to prevent default behavior.

4.1 Bank and Credit Market

After knowing their asset and entrepreneurial talent, individuals go to the bank and assess their financial situations. On the one hand, individuals save their money in the bank to earn interest. On the other hand, they can borrow money from the bank to start a firm. However, the credit market is not perfect due to the imperfect enforceability of contracts. Following Buera et al. (2011), I assume that individuals can renege on their contract and keep a $1 - s$ fraction of revenue net of labor cost. The only punishment is that they will lose the assets deposited in the bank.

The bank earns zero profit. The interest rate the bank pays is the same as the interest rate the bank lends. To further simplify the model, I assume people put all their assets in the bank and borrow the capital to start a firm. Under the model's assumption, this is equivalent to the case that adults use their assets to start a firm and save the rest in the bank if they have excess assets or borrow from the bank if they don't have sufficient capital.

If $s \geq \frac{\alpha}{1-\beta}$, the economy doesn't experience any financial friction, and the credit market is perfect, we are back to the benchmark model. When $0 \leq s < \frac{\alpha}{1-\beta}$, being aware that

individuals can default, the bank will set up a capital limit K_{it}^{limit} to prevent default. The bank wants to ensure that when utilized capital is smaller than or equal to K_{it}^{limit} , the default profit is always not larger than the profit of non-default. Mathematically, the bank sets up K_{it}^{limit} for each individual i_t such that:

$$\begin{aligned} & \max_{\{K_{it} \in [0, K_{it}^{limit}], L_{it}\}} (1-s) \{ (Az_{it})^{1-\alpha-\beta} (K_{it})^\alpha L_{it}^\beta - W_t L_{it} \} - a_{it} \\ & \leq \max_{\{K_{it} \in [0, K_{it}^{limit}], L_{it}\}} \{ (Az_{it})^{1-\alpha-\beta} (K_{it})^\alpha L_{it}^\beta - R_t K_{it} - W_t L_{it} \} + R_t a_{it} \end{aligned} \quad (17)$$

To make it easier to compare, I define $\tilde{s} \in [0, 1]$ where $s = \tilde{s} \times \frac{\alpha}{1-\beta}$.

Figure 3 and Figure 4 illustrate the capital limit for unbounded and bounded individuals separately. In both figures, the bank wants to set up K_{it}^{limit} such that the default profit is not larger than the production profit. The highest level of K_{it}^{limit} would be given by the intersection of π_{default} and $\pi_{\text{production}}$. In figure 3, the capital limit that the bank is willing to lend is higher than the individual's optimal capital level given by equation (8). So this individual will not be bound by the bank, and they can produce at the optimal capital level. In this case, I assume that the highest amount of capital the bank will lend would be K_{it}^* to simplify the model. However, in Figure 3, we can notice that $K_{it}^{limit} < K_{it}^*$. In this case, the capital individuals can borrow is less than the optimal amount they need. They can only produce at a suboptimal capital level if they become entrepreneurs. Individuals are unbounded if and only if:

$$\frac{a_{it}}{z_{it}} \geq A \times \frac{\alpha - s(1-\beta)}{1 + R_t} \left(\frac{\alpha}{R_t} \right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\alpha-\beta}} \quad (18)$$

In equilibrium, no one will default. What's more, K_{it}^{limit} is a function of the bank's seizure rate s , personal assets a_{it} and entrepreneurial ability z_{it} . $K_{it}^{limit}(s, a_{it}, z_{it})$ is increasing in s , a_{it} and z_{it} and concave in a_{it} and z_{it} .

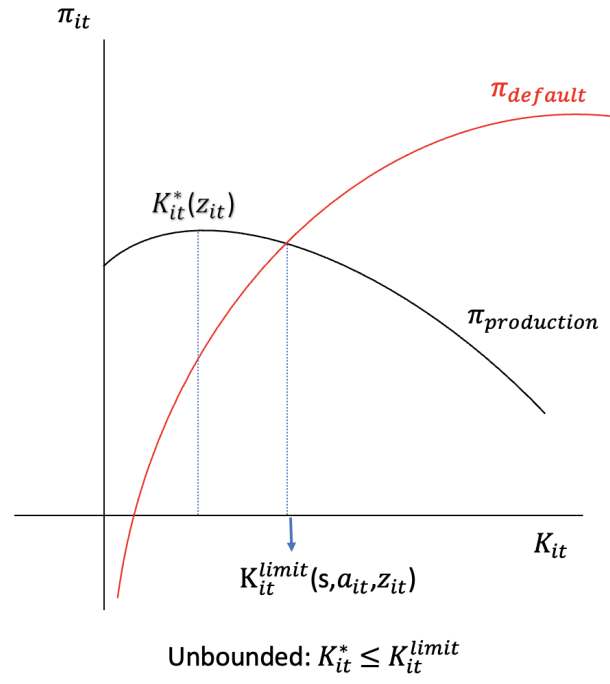


Figure 3: Capital Limit for Unbounded Individual

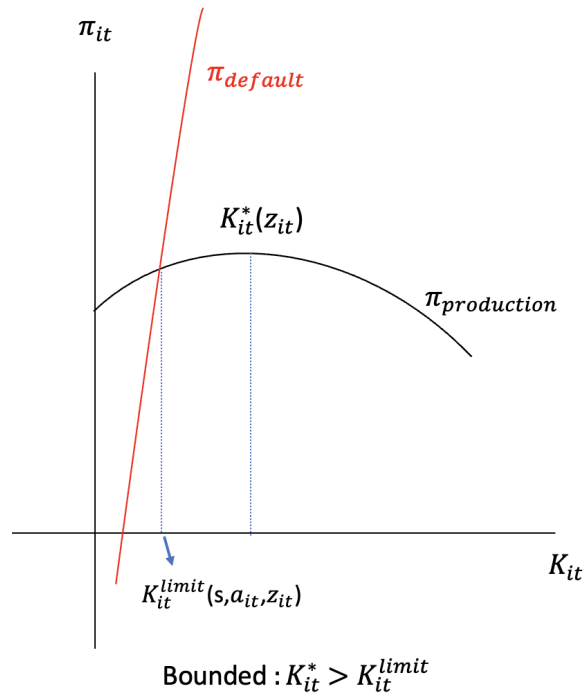


Figure 4: Capital Limit for Bounded Individual

4.2 Occupation, Income, and Utility

Individuals choose their occupations after knowing their capital limit K_{it}^{limit} from the bank.

They first face the following production problem with capital limit K_{it}^{limit}

$$\max_{\{K_{it} \leq K_{it}^{limit}, L_{it}\}} \pi_{it} = (Az_{it})^{1-\alpha-\beta} (K_{it})^\alpha L_{it}^\beta - R_t K_{it} - W_t L_{it} \quad (19)$$

For unbounded people, they will produce at the optimal capital level K_{it}^* based on equation (8) and earn profit π_{it}^* given in equation (10). For bounded individuals, they can only produce at a suboptimal level K_{it}^{limit} . Based on inequality (17), an individual's capital limit K_{it}^{limit} is given implicitly by equation (20). The production profit of bounded entrepreneurs would be (21).

$$s(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (K_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} - RK_{it}^{limit} + (1+R_t)a_{it} = 0 \quad (20)$$

$$\pi_{it}^B = (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} K_{it}^{limit \frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} (1-\beta) - R_t K_{it}^{limit} \quad (21)$$

As in the benchmark model, entrepreneurs can earn interest on their assets from the bank in addition to production profits. As a result, the total income of an entrepreneur would be:

$$y_{it}^E = \begin{cases} y_{it}^{UE} = \pi_{it}^* + R_t a_{it} & \text{if unbounded} \\ y_{it}^{BE} = \pi_{it}^B + R_t a_{it} & \text{if bounded} \end{cases} \quad (22)$$

The worker's income is the same as the benchmark model given by equation (12). Individuals choose the occupation that earns the highest income following equation (13).

For unbounded individuals, the entrepreneurial cutoff is the same as the benchmark model given by equation (14). For bounded individuals, their entrepreneurial cutoff will be decided by equation (23). Bounded individuals with entrepreneurial ability higher than or equal to z_{it}^{BE} become bounded entrepreneurs.

$$z_{it}^{BE} = \frac{1}{A} \left(\frac{W_t + R_t K_{it}^{limit}}{(1 - \beta)(K_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}} \right)^{\frac{1-\beta}{1-\alpha-\beta}} \quad (23)$$

Figure 5 shows individuals' occupational choices under financial frictions. The Y-axis shows the asset level, while the X-axis presents the entrepreneurial talent. The orange line indicates the bound status given by equation (18). Individuals above the line are unbounded, while those under the line are bounded. The vertical blue line is the entrepreneurial cutoff for unbounded individuals given in equation (14). Unbounded individuals with talent equal to or higher than the cutoff will become entrepreneurs. The green curve line demonstrates the entrepreneurial cutoff for bounded people according to equation (23). Bounded people above the line choose to become bounded entrepreneurs.

One comparative static analysis we could do here is to check the influence of the bank's seizure rate s while fixing R and W . In a model without financial frictions, individuals with lower entrepreneurial ability choose to become workers because the labor income is higher than entrepreneurial income: $y_{it}^W > y_{it}^{UE}$. When we add financial frictions, we know that bounded entrepreneurial income must be no greater than unbounded entrepreneurial income: $y_{it}^{UE} \geq y_{it}^{BE}$. As a result, we should have $y_{it}^W > y_{it}^{UE} \geq y_{it}^{BE}$. Individuals left to the unbounded entrepreneurial cutoff still choose to become workers in a world with financial frictions. For those who are entrepreneurs in a frictionless world, there are three situations. Those with small amount of assets now become workers since $y_{it}^{UE} \geq y_{it}^W > y_{it}^{BE}$. Individuals with middle-level of assets now become bounded entrepreneurs because $y_{it}^{UE} \geq y_{it}^{BE} > y_{it}^W$. Finally, individuals with the highest level of assets remain unbounded entrepreneurs. Comparing

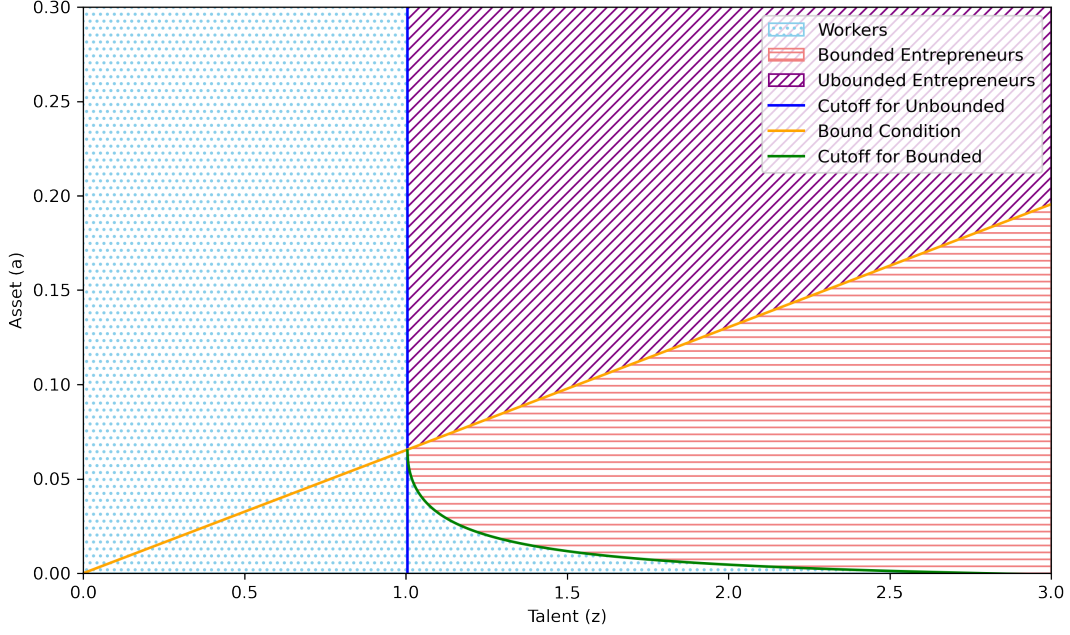


Figure 5: Occupation Choices in the Model with Frictions

the frictionless model with the model considering financial frictions, we can see occupation distortions and mismatches between entrepreneurial talent and capital.

After knowing the occupation and corresponding income, an individual maximizes their utility according to (1). The optimal decision rules are precisely the same as those in the benchmark model listed in equation (2), equation (3), and equation (4).

4.3 Equilibrium

We have $\mathbf{x} = (a, z)$ as the state variable. We can calculate $P(\mathbf{x}'|\mathbf{x})$ as the transition function. In equilibrium, given the bank's seizure rate s , we should have a invariant distribution of the state variable $d^*(\mathbf{x})$; optimal solutions to utility maximization problem $c(\mathbf{x})$, $n(\mathbf{x})$ and $m(\mathbf{x})$; occupational choice $o(\mathbf{x})$; bound status of individuals $B(\mathbf{x})$ and corresponding capital and labor usage $K(\mathbf{x})$ and $L(\mathbf{x})$; interest rate R and wage W . In equilibrium, we should have:

- (a) Adults maximize utility subject to the budget constraint.
- (b) The bank sets up capital limits for each individual solving (17).
- (c) Entrepreneurs maximize profits given bound status and K_{it}^{limit} .
- (d) Adults make optimal occupational choices based on (13).
- (e) Labor and capital markets clear:

$$\int_{i_t \in W} di_t = \int_{i_t \in E \& Unbounded} L_{it}^* di_t + \int_{i_t \in E \& Bounded} L_{it}^{limit} di_t \quad (24)$$

$$\int_{i_t} a_{it} di_t = \int_{i_t \in E \& Unbounded} K_{it}^* di_t + \int_{i_t \in E \& Bounded} K_{it}^{limit} di_t \quad (25)$$

- (f) d^* is stable.

5 Calibration

In this section, I use PSID (The Panel Study of Income Dynamics) data to calibrate the model's parameters under financial frictions. I will first describe the calibration strategy and then show details about the calibration method and results.

5.1 Calibration Strategy Overview

The inputs of the calibration would be the asset and talent distributions. I estimate these distributions based on the PSID dataset. After knowing the talent and asset distributions, I first calibrate four macro-level parameters: the TFP shifter A , the capital share α , the labor share β , and the seizure rate s . The targeted moments would be the entrepreneurship rate, the top 5% income share, the log mean of income, and the log median of income.

After calibrating the macro-level parameters, I use the estimated parameter values and the distribution of talent and assets to generate the estimated income. Then I calibrate the utility parameters using the estimated income to target the log mean of consumption and the histogram of fertility. The estimated utility parameters includes: consumption weight θ_1 , bequest weight θ_2 , elasticity ϵ and time cost per child τ .

5.2 PSID Data

PSID is longitudinal data from a representative sample of U.S. households. It includes intergenerational data on income, consumption, fertility, and occupation over a long period. Before 1999, the survey was conducted annually. After the 1999 wave, it switched to a biennial cycle. My sample consists of the 1968 to 2011 waves of the PSID and keeps the data for the household head. However, the consumption data regarding detailed categories is not included until the 1999 wave. To extend the dataset, I use the consumption data from Attanasio and Pistaferri (2014), which uses socioeconomic variables, prices, and food consumption, which are consistently available in the PSID, to impute the consumption data before 1999.

One concern is that although PSID provides income and consumption data at the annual or biennial level, some years may be missing. It may not contain an individual's income and consumption data throughout their life cycle. Therefore, I estimate the income and consumption data for one particular year using the following equation. For y_{it} and c_{it} :

$$\ln y_{it} = \beta_0 + \beta_1 \times age_{it} + \omega_i^y \quad (26)$$

$$\ln c_{it} = \gamma_0 + \gamma_1 \times age_{it} + \omega_i^c \quad (27)$$

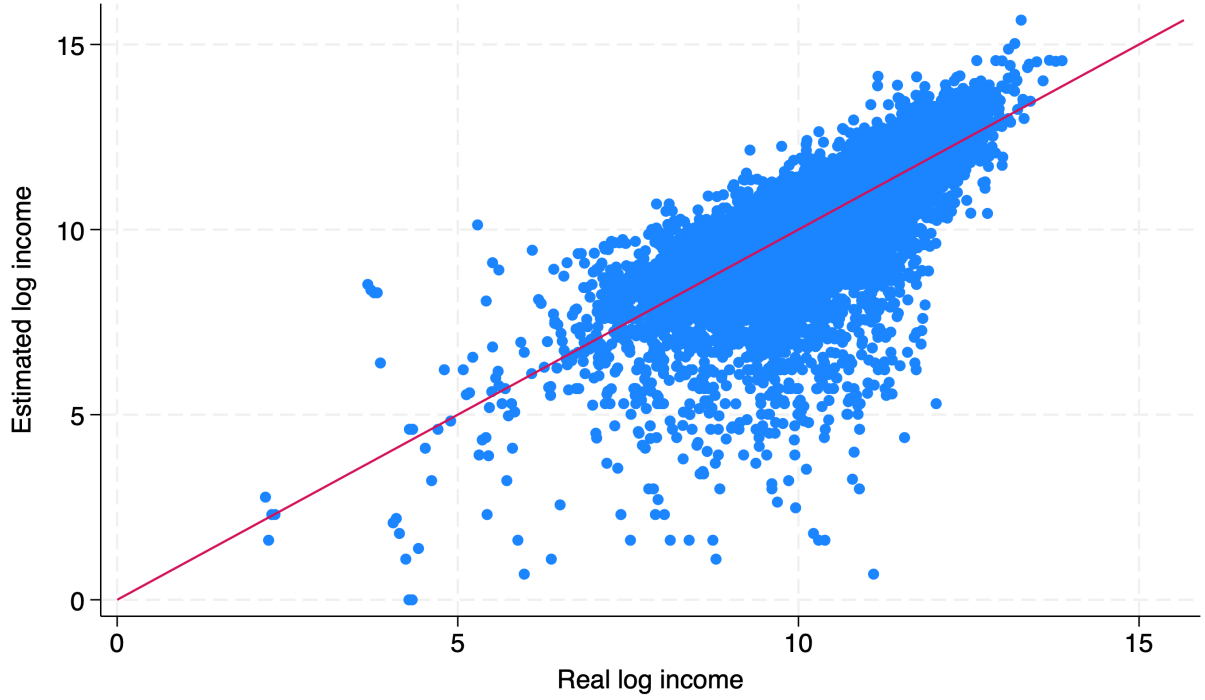


Figure 6: Real log income and estimated log income

t indicates the year when the individuals are aged between 25 and 65. ω_i^y and ω_i^c are the individual fixed effects for income and consumption, which also capture cohort fixed effects. After running the regression, I store the individual fixed effects and estimate yearly income and consumption using the regression coefficients. To get the lifetime income, I use a constant interest rate $R = 5\%$ to discount all the yearly incomes from age 25 to age 65. I use the same method to calculate the expected lifetime consumption.

Figure 6 shows the estimated and real log income levels. Similarly, Figure 7 shows the relationship between the estimated and real log consumption.

In my model, the asset level refers to the bequest inherited from the parent. Since the bequest data from the PSID contains a lot of missing values, I define the inherited asset as the asset level at age 25 and estimate it using the following equation:

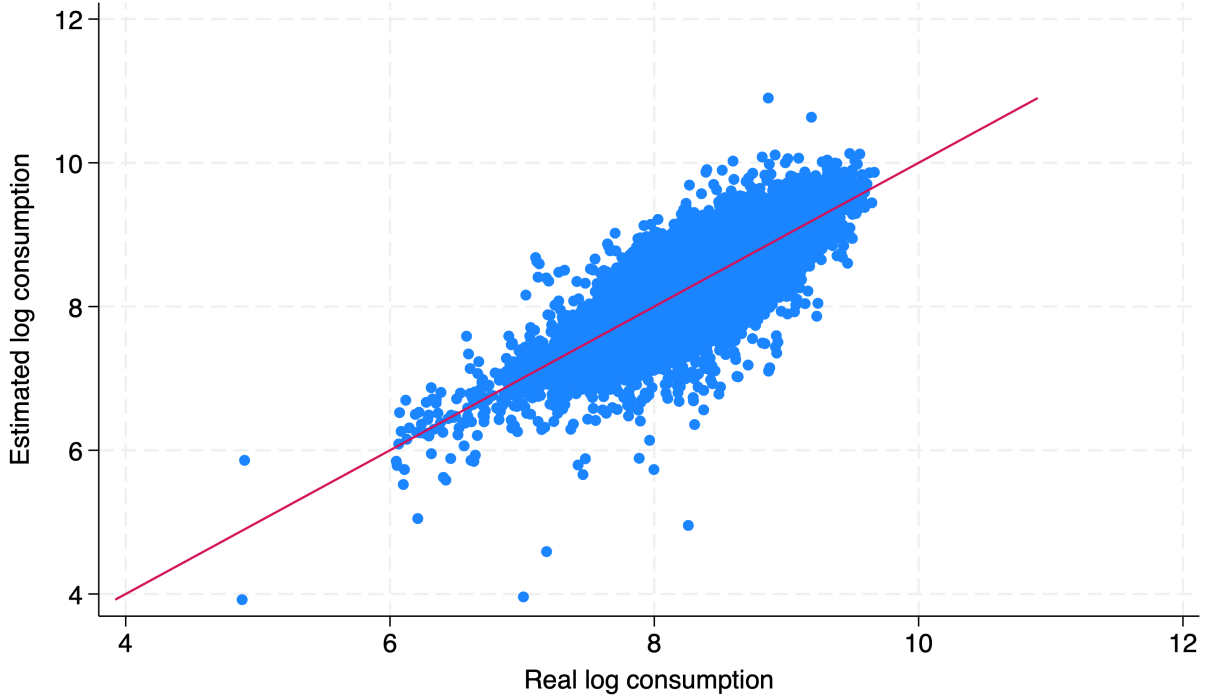


Figure 7: Real log consumption and estimated log consumption.

$$a_{it} = \sigma_0 + \sigma_1 \times \text{age} + \omega_i^a \quad (28)$$

ω_i^a indicates the individual fixed effect, which also captures the cohort fixed effect. Figure 8 shows the relationship between estimated and real log asset levels. After calculating the lifetime income, lifetime consumption, and asset level at age 25, I discount all the values back to the year 1980 using the interest rate $R = 5\%$.

Another important point to discuss is how to identify entrepreneurs in the dataset. I use two variables in the PSID dataset. The first variable relates to whether the household head is reported as self-employed. This question could capture whether the individual is working for someone else or themselves. The second variable regards whether the family owns a business or has a financial interest in any business enterprise. This question provides information about the ownership of a business. If the answers to both questions are yes, I define the

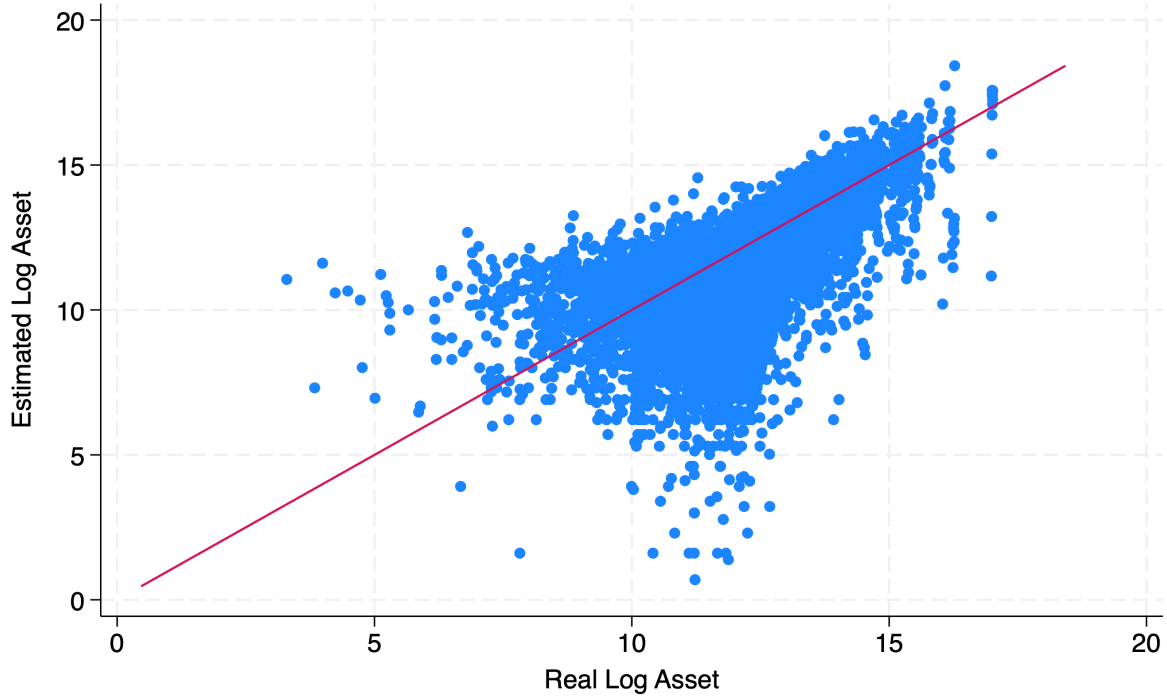


Figure 8: Real log asset and estimated log asset

individual as an entrepreneur in that particular year. If a person is an entrepreneur for two consecutive years and for at least half of the years observed in the PSID dataset, I define them as an entrepreneur.

5.3 Asset and Talent Distributions

Following the literature, I assume the asset follows the lognormal distribution. The asset distribution is based on the estimated asset level at age 25 based on the PSID dataset. Figure 9 shows the estimated and real asset distributions.

As for the talent distribution, I first use the following equation to get the residues of lifetime income as the talent measure:

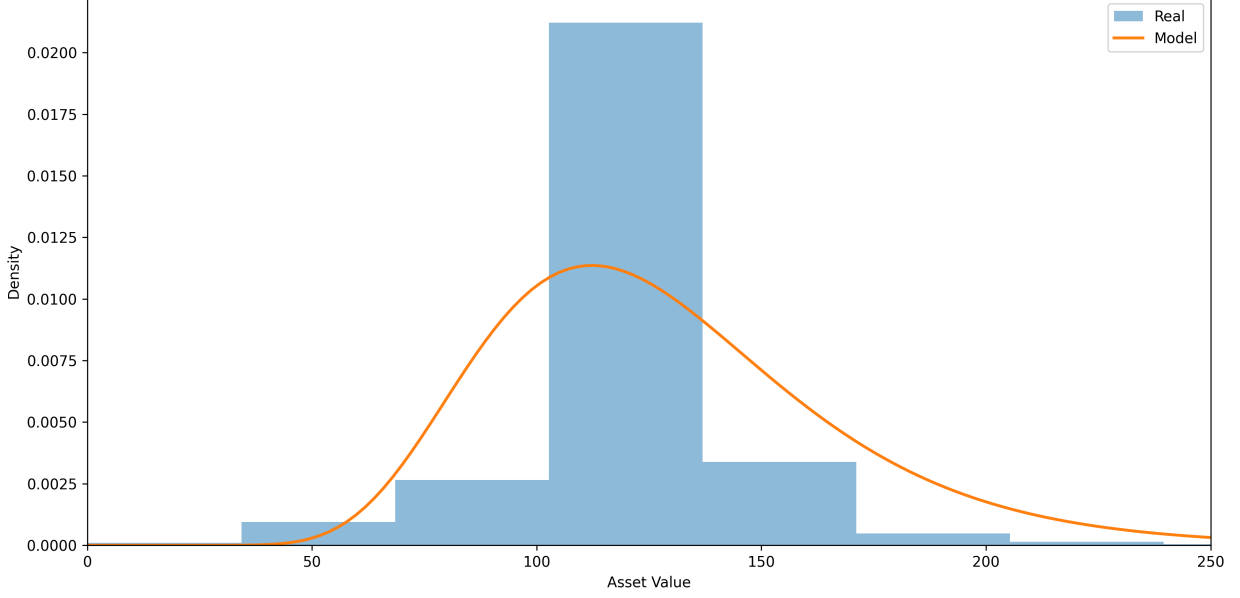


Figure 9: Asset Distribution (in units of \$100,000)

$$\begin{aligned}
\ln(y_i) = & \omega_0 + \omega_1^\top \text{asset_quantile}_i + \omega_2^\top \text{race}_i + \omega_3^\top \text{disability}_i \\
& + \omega_4^\top \text{gender}_i + \omega_5^\top \text{birthyear}_i \\
& + \omega_6^\top (\text{asset_quantile}_i \odot \text{race}_i) + \omega_7^\top (\text{asset_quantile}_i \odot \text{disability}_i) \\
& + \omega_8^\top (\text{asset_quantile}_i \odot \text{gender}_i) + \omega_9^\top (\text{asset_quantile}_i \odot \text{birthyear}_i) + \varepsilon_i
\end{aligned} \tag{29}$$

In the equation, **asset_quantile**_{*i*} indicates the wealth quantile of the individual in the society, and **disability**_{*i*} indicates whether the individual is handicapped. I treat the residue as the log value of the talent.

I need to estimate both the talent distribution and the AR(1) process parameters. For the initial distribution, I use the log-normal distribution to simulate, but assume there exists a small fraction of super-talented individuals on the far right end. Figure 10 shows the comparison between the real data and simulated talent distribution.

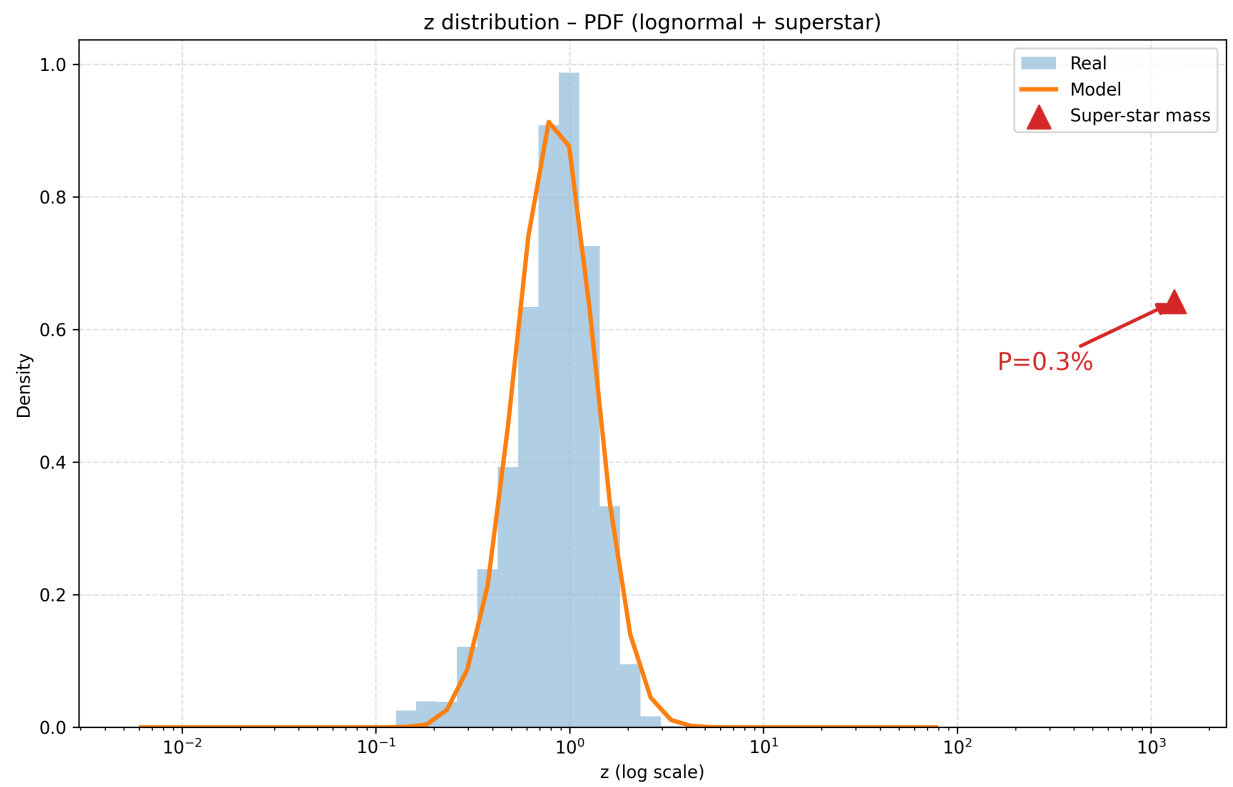


Figure 10: Talent distribution

I assume the intergenerational talent transmission follows the AR(1) process. Since the dataset only includes observations for the household head, and most of the households are male, I link the relationship between the father and the children and run the regression following equation (30). The dataset contains 746 matched father–child pairs.

$$\ln z_{it+1} = \bar{z} + \rho \ln z_{it} + \epsilon_{it} \quad (30)$$

After running the regression in equation (30), I get the parameter accordingly. Table 1 lists the estimated parameters for the AR(1) process of z .

Table 1: Estimated Parameters for Talent Inheritance

Parameter	Description	Value
ρ	Persistence of talent	0.29
$\bar{z}$	Mean of log-talent	-0.1
$\delta_{\epsilon_{it}}$	Std. dev. of talent shock	0.43

Notes: This table reports the estimated parameters governing the stochastic process of talent inheritance. ρ represents intergenerational persistence, $\bar{z}$ is the mean of log talent, and $\delta_{\epsilon_{it}}$ is the standard deviation of idiosyncratic talent shocks.

5.4 Macro Parameters

After estimating the distribution of assets and talent, I have four parameters to be estimated at the macro level: the TFP shifter A , the capital share α , the labor share β , and the bank’s seizure rate s . The four main targets are the fraction of entrepreneurs within the U.S., the top 5% income share, the log mean of income, and the log median of income.

5.5 Utility Parameters

After calibrating the macro-level parameters, I use the estimated income to calibrate the following four parameters in the utility maximization problem: the weight of consumption θ_1 , the weight of total bequest θ_2 , the elasticity of substitution ϵ , and the time cost per child τ .

From the solution to the utility maximization problem in (1), we know the optimal decisions for consumption and fertility. I use GMM (General Method of Moments) to target the following moments: the log mean of consumption and the distribution of the number of children.

5.6 Calibration Results

Table 2 shows the calibrated parameter values. For the macro-level parameters, the standardized bank seizure rate $\tilde{s}$ is very close to 1, indicating that the US's financial market is near perfect. The result shows that most entrepreneurs in the US are not bound by the bank unless they have high entrepreneurial talent and need to open very large firms, or they have a large amount of wealth but low entrepreneurial talent.

As for the utility parameters, the consumption weight θ_1 is around 0.13 and the total bequest weight θ_2 is around 0.22. We can observe that the elasticity $\epsilon = 4.49$, which is larger than 1, indicating that the number of children and the income level exhibit a hump-shaped relationship. The time cost of raising one child is 0.19.

Table 3 Panel A lists the comparison between the model's predictions and real data for the targeted moments. For all the targeted moments, the model fits the real data well.

Figure 11 represents the comparison of income between the model's prediction and the

Table 2: Calibrated Parameters

Parameter	Description	Value
<i>Panel A: Financial Frictions</i>		
$\tilde{s}$	Standardized seizure rate	0.98
<i>Panel B: Production</i>		
A	TFP shifter	136
α	Capital share	0.62
β	Labor share	0.28
<i>Panel C: Preferences</i>		
θ_1	Consumption weight	0.13
θ_2	Bequest weight	0.22
ϵ	Elasticity of substitution	4.49
τ	Per-child time cost	0.19

Notes: This table reports the calibrated parameters for the model with financial frictions. Panel A indicates the financial friction parameter. Panel B shows the parameters in the production function. Panel C presents the parameters in the utility maximization problem.

Table 3: Model Fit: Targeted and Untargeted Moments

Moment	Data	Model
<i>Panel A: Targeted Moments</i>		
Entrepreneurship rate	0.099	0.097
Top 5% share	0.14	0.12
Log mean income	3.55	3.37
Log median income	3.43	3.29
Log mean consumption	0.85	0.62
<i>Panel B: Untargeted Moments (Validation)</i>		
Entrep. rate: Low-income	0.057	0.053
Entrep. rate: Middle-income	0.073	0.088
Entrep. rate: High-income	0.168	0.15
Entrep./Worker income ratio	1.469	1.661

Notes: This table compares the moments in the data with those generated by the model. Panel A reports the moments used in the calibration. Panel B reports untargeted moments. The model also targets the distribution of n . See the comparison of fertility distribution in Figure 13.

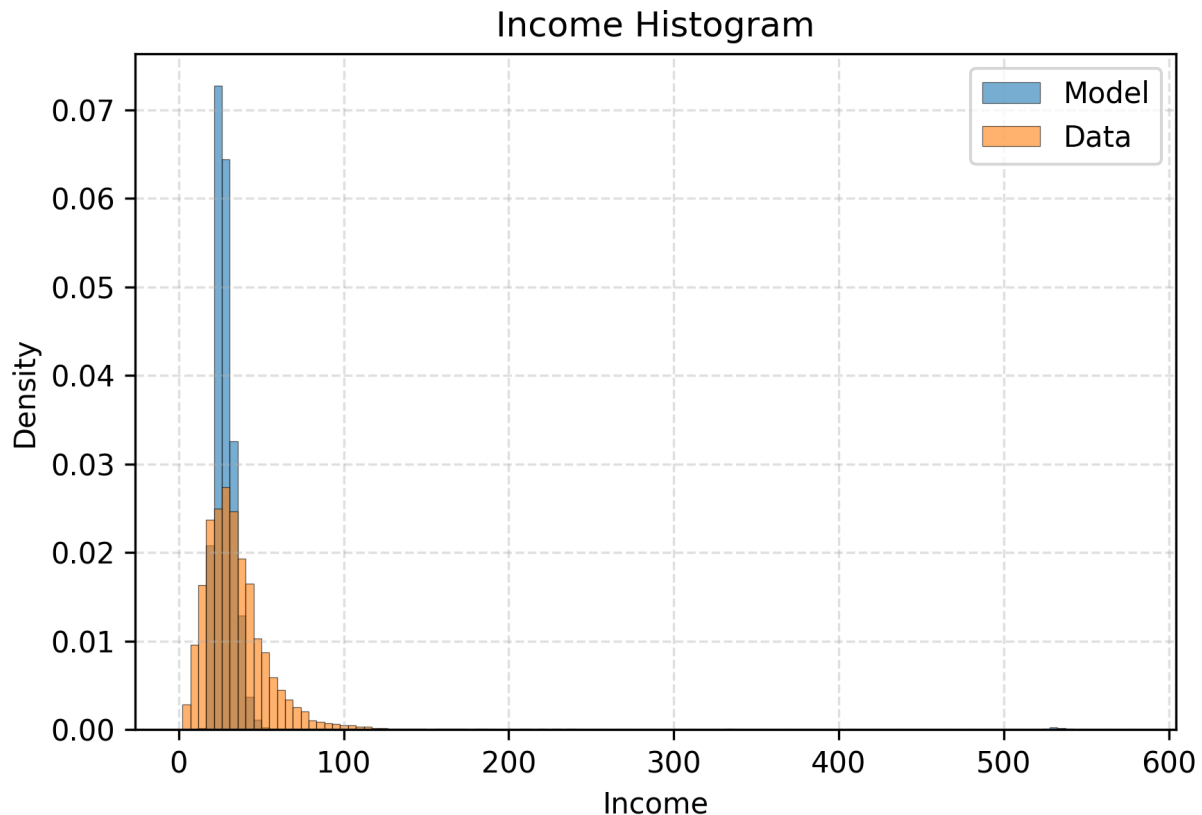


Figure 11: Income: Data vs Model

real data, while Figure 12 and 13 represent the comparison of consumption and the number of children, respectively. For the income and the number of children, the model predicts the distributions very well. For consumption, even though I have only targeted the log mean of the consumption, the model predicts the overall distribution of consumption very well.

Finally, Table 3, Panel B, compares the model's predictions with empirical data for untargeted moments. The first three rows report entrepreneurship ratios for low-income, middle-income, and high-income individuals. The last row compares the income ratio between entrepreneurs and workers. In all cases, the model's predictions match the empirical data well.

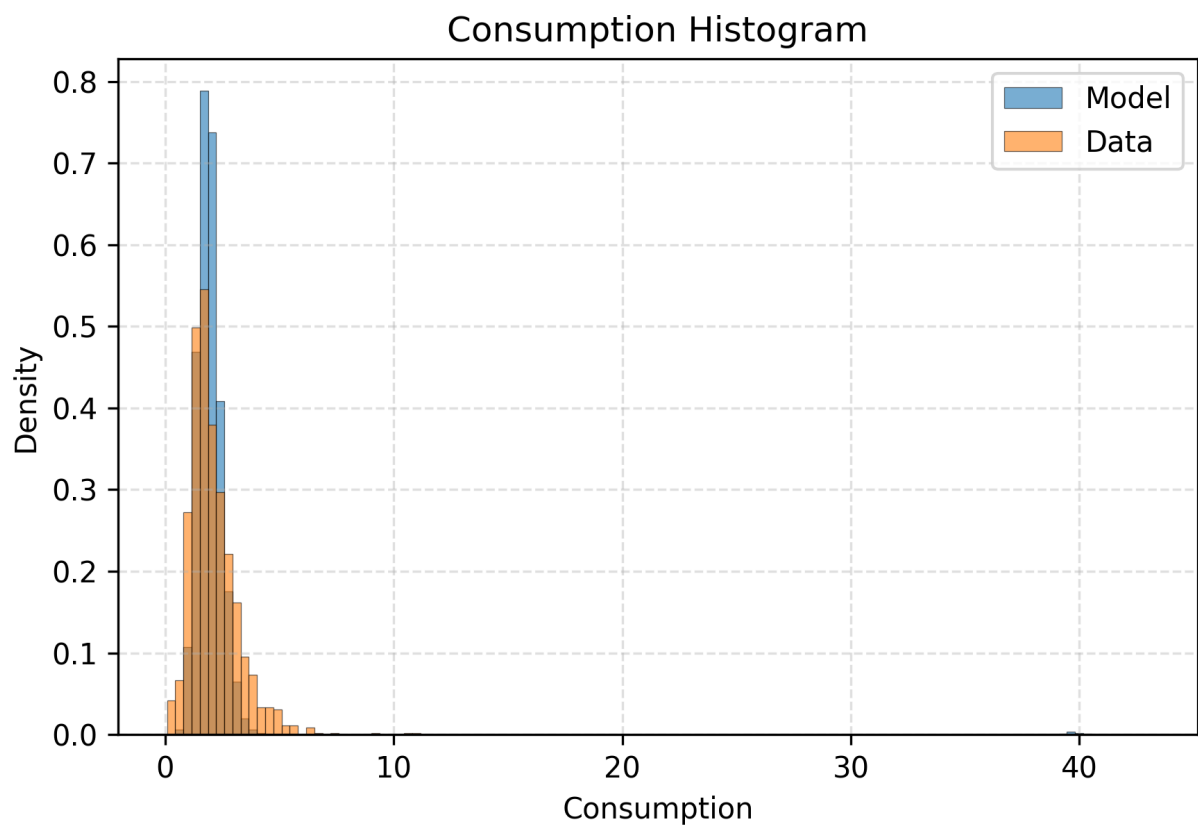


Figure 12: Consumption: Data vs Model

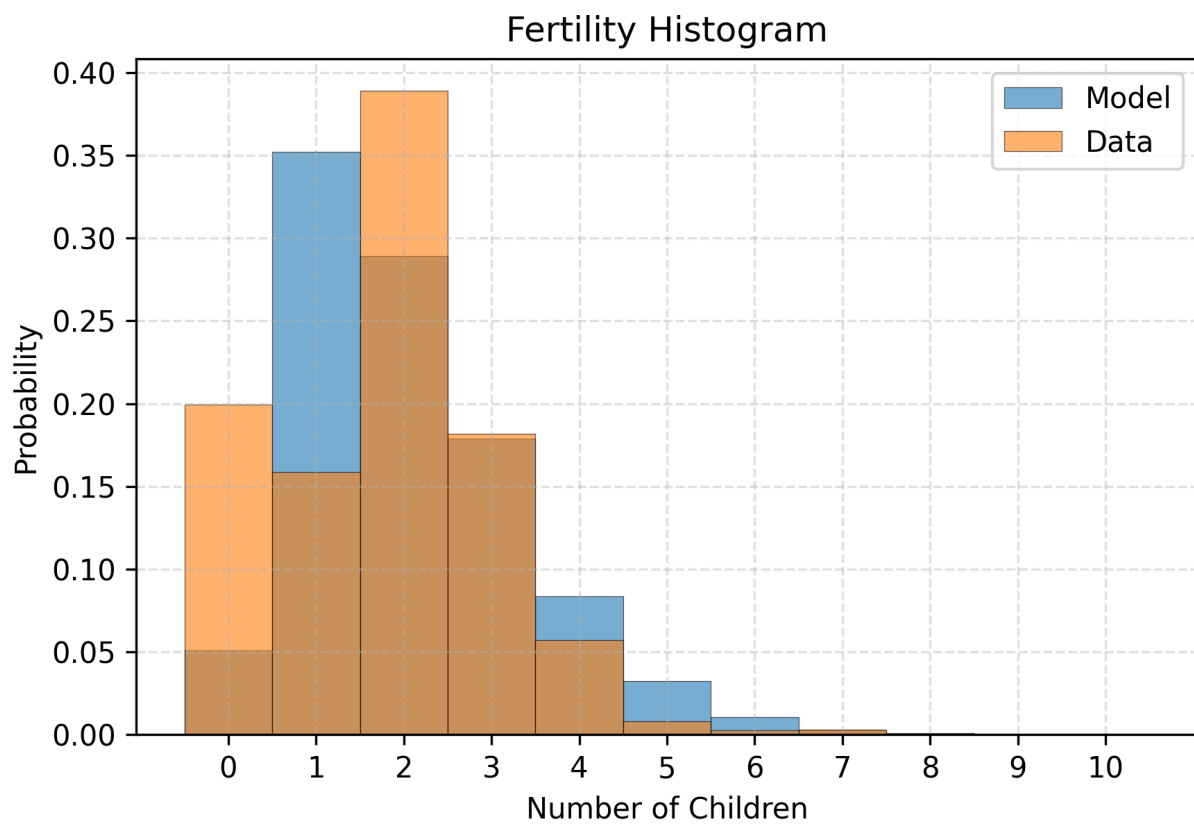


Figure 13: Fertility: Data vs Model (Model predictions round to integers)

6 Counterfactual Analysis

In this section, I present the counterfactual analysis for changing the financial frictions and shutting down the fertility channel. I examine how the entrepreneurship rate, income inequality, mobility, economic development, and social welfare change in the steady state under different conditions.

Table 4 shows the results for all the counterfactual analyses. As for entrepreneurship, both an increase and a decrease in financial frictions relative to the baseline model would decrease the entrepreneurship rate in the steady state. The difference in entrepreneurship rate between Table 4 and Table 3 is due to the fact that Table 4 indicates the entrepreneurship rate in the long run. What's more, if I shut down the fertility channel, we would overestimate the entrepreneurship rate.

As for income inequality, we can compare the results of the Gini coefficient and the top 5% income share. We can see similar patterns in both indicators that both increases and decreases in financial frictions would lead to greater income inequality compared with the baseline model. Shutting down the fertility channel would overestimate income inequality, except in the case that when $\tilde{s} = 0$ and forcing $n = 1$ would underestimate the Gini coefficient.

Intergenerational mobility is defined as the probability that a child's income quantile is higher than that of the parent. If I increase or decrease the financial frictions relative to the baseline model, we would see a drop in mobility. If I shut down the fertility channel, we would see an overestimation of mobility, except when $\tilde{s} = 0$.

It may seem counterintuitive that, when the financial market improves relative to the baseline model, entrepreneurship drops slightly and equality and mobility decrease. The mechanism is that reducing the financial barriers from an already well-functioning financial

Table 4: Counterfactual Analysis: Financial Frictions and Fertility

Moment	Endogenous Fertility			Fixed Fertility ($n = 1$)		
	s^{base}	$\tilde{s} = 1$	$\tilde{s} = 0$	s^{base}	$\tilde{s} = 1$	$\tilde{s} = 0$
<i>Panel A: Macroeconomic Aggregates</i>						
Entrepreneurship rate	0.216	0.200	0.114	0.218	0.214	0.170
Income per capita	0.457	0.523	0.391	0.962	0.976	0.267
Consumption	0.012	0.013	0.009	0.043	0.044	0.012
<i>Panel B: Inequality and Mobility</i>						
Gini coefficient	0.560	0.609	0.643	0.623	0.627	0.634
Top 5% income share	0.555	0.594	0.630	0.632	0.637	0.641
Intergen. mobility	0.119	0.115	0.0009	0.167	0.152	0.0002
<i>Panel C: Welfare and Demographics</i>						
Utility	0.204	0.236	0.184	0.686	0.687	0.625
Fertility (n)	0.299	0.345	0.274	–	–	–

Notes: This table summarizes the counterfactual analysis across different levels of financial frictions ($\tilde{s}$) and fertility setups. The first three columns report results under the model with endogenous fertility. The last three columns report results for an economy where fertility is exogenously fixed at $n = 1$.

market allows highly talented individuals to absorb more capital and drive up interest rates. Consequently, less productive firms are pushed out of the market, the return to wealth increases, leading to fewer entrepreneurs, higher inequality, and lower mobility.

Decreasing financial frictions would increase income per capita, consumption, and utility. If I force $n = 1$, we would overestimate consumption and utility. As for income per capita, forcing $n = 1$ would overestimate it for the baseline model and for $\tilde{s} = 1$. However, in the collateral case when $\tilde{s} = 0$, excluding the fertility channel would underestimate income per capita.

Finally, I want to check the influence of financial frictions on fertility. Reducing financial frictions would increase fertility.

7 Conclusion

This paper examines how income-driven parental choices regarding fertility and bequests impact children's entry to entrepreneurship under financial frictions. To study these dynamics, I propose an OLG model with endogenous fertility, bequest, and occupational choices, and calibrate the model using US data. The model indicates that the interaction between parental decisions and borrowing constraints could lead to the misallocation of entrepreneurial talent and capital. Counterfactual analysis indicates that reducing financial frictions relative to the baseline model would increase income per capita but slightly reduce entrepreneurship due to the increased interest rate, thereby increasing inequality and reducing mobility. These findings underscore the importance of designing financial policies that promote efficiency while also addressing concerns about income concentration.

While the model allows for intergenerational persistence in entrepreneurial talent through an AR(1) process, it abstracts from endogenous parental human capital investment decisions

that may also shape children's entrepreneurship potentials. Extending the framework to incorporate such investment choices would be a useful direction for future research and could further clarify how financial and human capital channels jointly affect entrepreneurship and intergenerational mobility.

In addition, although the quantitative analysis is calibrated to the U.S. economy, the mechanism studied here may be even more salient in countries at earlier stages of financial development, where entrepreneurial entry depends more heavily on family resources. Extending the analysis to such settings would be another useful direction for future research and could shed further light on how family resources shape entrepreneurship and intergenerational mobility differently across different stages of economic and financial development.

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Appendix

A Theoretical Proof

A.1 Benchmark Model without Financial Frictions

A.1.1 The Entrepreneur's Problem

Assume the depreciation rate is 1. After each time period, all of the capital borrowed will be utilized and won't be transferred to the next period. As an entrepreneur, the individual needs to solve the following problem in the per capita setup:

$$\max_{k_{it}, l_{it}} \pi_{it} = (Az_{it})^{1-\alpha-\beta} (k_{it})^\alpha l_{it}^\beta - R_t k_{it} - W_t l_{it}$$

z_{it} is the entrepreneurial talent of the individual. The first-order conditions of the problem are given by:

$$(Az_{it})^{1-\alpha-\beta} \alpha (k_{it})^{\alpha-1} l_{it}^\beta = R_t \tag{31}$$

$$(Az_{it})^{1-\alpha-\beta} (k_{it})^\alpha \beta l_{it}^{\beta-1} = W_t \tag{32}$$

$$\frac{k_{it}}{l_{it}} = \frac{\alpha W_t}{\beta R_t} \tag{33}$$

The optimal choices are:

$$k_{it}^* = \left(\frac{\alpha}{R_t}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} Az_{it} \quad (34)$$

$$l_{it}^* = \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{1-\alpha}{1-\alpha-\beta}} Az_{it} \quad (35)$$

The maximized firm profit is:

$$Az_{it} \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} (1 - \alpha - \beta)$$

A.1.2 Entrepreneurship Decision

After evaluating the profit of starting a business, the individual will then try to compare the income of an entrepreneur and a worker to choose the occupation that will bring the highest income. The household chooses to start a firm if and only if the income from being an entrepreneur exceeds the income from working:

$$Az_{it} \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} (1 - \alpha - \beta) + R_t a_{it} \geq W_t + R_t a_{it}$$

After solving the problem, we get the entrepreneurial cutoff. The individual only chooses to become an entrepreneur if their talent is higher than a threshold.

$$Az_{it} \geq \frac{\beta}{1 - \alpha - \beta} \left(\frac{R_t}{\alpha}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{W_t}{\beta}\right)^{\frac{1-\alpha}{1-\alpha-\beta}} \quad (36)$$

Define the entrepreneurial threshold as z_t^e . Note that in the benchmark model, the cutoff is universal across individuals.

Let's also define the Probability Density Function and Cumulative Distribution Function of entrepreneurial talent as $g(z)$ and $G(z)$, respectively. Moreover, I also define the average talent of entrepreneurs as $\phi(z)$:

$$\phi(z) = \int_{z_{it} \geq z_t^e} z_{it} g(z_{it}) dz_{it}$$

Now we have:

$$\frac{1}{N_t} \int_{i_t \in E} z_{it} d_{i_t} = E(z_{it} | z_{it} \geq z_t^e) \times (1 - G(z_t^e)) = \int_{z_{it} \geq z_t^e} z_{it} g(z_{it}) dz_{it} = \phi_t \quad (37)$$

A.1.3 Labor Market

From equation (35) and (37), we can calculate the labor demand per capita as follows:

$$l_t^d = \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{1-\alpha}{1-\alpha-\beta}} A \phi_t \quad (38)$$

As for labor supply per capita, I assume every adult worker provides one unit of labor inelastically. The market-clearing condition would be:

$$1 = \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{1-\alpha}{1-\alpha-\beta}} A \phi_t \quad (39)$$

A.1.4 Capital Market

According to equation (34), capital demand per capita is given by:

$$\int_{i_t \in \{E\}} k_{it}^s df(a_{it}) = \left(\frac{\alpha}{R_t}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} A \phi_t \quad (40)$$

The next step is to calculate the capital supply:

$$k_t^s = \int_{i_t} a_{it} df(a_{it}) \quad (41)$$

The clearing condition for the capital market is:

$$\int_{i_t} a_{it} f(a_{it}) = \left(\frac{\alpha}{R_t}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} A \phi_t \quad (42)$$

A.1.5 Equilibrium

In the benchmark model, we have the market-clearing condition for labor and capital from equation (39) and equation (42) as well as the relationship between R_t and W_t from equation (33). So now we have:

$$\frac{\alpha W_t}{\beta R_t} = \frac{\int_{i_t} a_{it} df(a_{it})}{G(z_t^e) N_t} = \frac{\bar{k}_t}{G(z_t^e)} \quad (43)$$

$$R_t = \alpha \left(\frac{1}{\bar{k}_t}\right)^{1-\alpha} (G_t)^\beta (A \phi_t)^{1-\alpha-\beta} \quad (44)$$

$$W_t = \beta(k_t)^\alpha \left(\frac{1}{G_t}\right)^{1-\beta} (A\phi_t)^{1-\alpha-\beta} \quad (45)$$

k_t is the equilibrium capital per capita level.

A.1.6 Solution Existence & Uniqueness of z_t^e

Now using equation (44) and equation (45) we can rewrite the entrepreneurial cutoff as follows:

$$z_t^e = \frac{\beta}{1 - \alpha - \beta} \frac{\phi_t}{G_t} \quad (46)$$

Define:

$$\begin{aligned} f(z_t^e) &= \beta \frac{\phi_t}{G_t} \\ j(z_t^e) &= (1 - \alpha - \beta) z_t^e \end{aligned}$$

We need to show that there is one unique z_t^e which could solve the following equation:

$$j(z_t^e) = f(z_t^e) \quad (47)$$

It's easy to show that:

$$\phi_t'(z_t^e) < 0$$

$$G_t'(z_t^e) > 0$$

As a result $f(z_t^e)$ is decreasing in z_t^e and $j(z_t^e)$ is increasing in z_t^e . When $z_t^e \rightarrow 0$, $G_t \rightarrow 0$, ϕ_t converges to a constant, $f(z_t^e)$ converges to infinity. When $z_t^e \rightarrow \infty$, $G_t \rightarrow 1$, $\phi_t \rightarrow 0$, $f(z_t^e) \rightarrow 0$. According to the Intermediate Value Theorem, there must be one unique z_t^e which could make equation (47) hold.

A.1.7 Law of Motion for Capital

Next, consider the law of motion for capital. According to the solution of the utility maximization problem, we know the optimal solution to total bequest m_{it} and fertility n_{it} are given by:

$$m_{it}^* = \frac{\left(\frac{\theta_2}{\theta_1}\right)^\epsilon}{1 + \left(\frac{\theta_2}{\theta_1}\right)^\epsilon + \left(\frac{1-\theta_1-\theta_2}{\theta_1\tau y_{it}}\right)^\epsilon} \times y_{it}$$

$$n_{it}^* = \frac{\left(\frac{1-\theta_1-\theta_2}{\theta_1\tau}\right)^\epsilon}{1 + \left(\frac{\theta_2}{\theta_1}\right)^\epsilon + \left(\frac{1-\theta_1-\theta_2}{\theta_1\tau y_{it}}\right)^\epsilon} \times y_{it}^{1-\epsilon}$$

Accordingly, the capital per capita next generation should be defined as:

$$k_{t+1} = \frac{\int_{i_t} m_{it} di_t}{\int_{i_t} n_{it} di_t} = \frac{\int_{i_t} \frac{\left(\frac{\theta_2}{\theta_1}\right)^\epsilon}{1 + \left(\frac{\theta_2}{\theta_1}\right)^\epsilon + \left(\frac{1-\theta_1-\theta_2}{\theta_1\tau y_{it}}\right)^\epsilon} \times y_{it} di_t}{\int_{i_t} \frac{\left(\frac{1-\theta_1-\theta_2}{\theta_1\tau}\right)^\epsilon}{1 + \left(\frac{\theta_2}{\theta_1}\right)^\epsilon + \left(\frac{1-\theta_1-\theta_2}{\theta_1\tau y_{it}}\right)^\epsilon} \times y_{it}^{1-\epsilon} di_t} \quad (48)$$

The capital per capita of the next generation is purely decided by the income distribution of the current generation.

A.2 The Model with Financial Frictions

A.2.1 The Bank's Problem

First, we need to introduce the bank's problem when individuals may default after borrowing money from the bank. Banks are in a competitive market. They aim to sell as much capital as possible without making the entrepreneurs default.

$$\max_{k_{it}} R_t \left(\int_{i_t \in E} k_{it} d_{it} - \int_{i_t} a_{it} d_{it} \right)$$

However, given the imperfect contract enforceability determined by the parameter s , entrepreneurs can renege on their contracts and keep a fraction $(1 - s)$ of the profit without paying the capital interest. The bank sets the capital limit to prevent entrepreneurs from defaulting. Entrepreneurs don't default if and only if:

$$\begin{aligned} & \max_{k_{it} \in [0, k_{it}^{limit}]} (1 - s) \{ (Az_{it})^{1-\alpha-\beta} (k_{it})^\alpha l_{it}^\beta - W_t l_{it} \} - a_{it} \\ & \leq \max_{k_{it} \in [0, k_{it}^{limit}]} \{ Az_{it}^{1-\alpha-\beta} (k_{it})^\alpha l_{it}^\beta - R_t k_{it} - W_t l_{it} \} + R_t a_{it} \end{aligned} \quad (49)$$

We know that from firm's problem, given any k_{it} , the optimal level of labor should be:

$$\bar{l}_{it} = \left(\frac{\beta z_{it}^{1-\alpha-\beta} (k_{it})^\alpha}{W_t} \right)^{\frac{1}{1-\beta}} \quad (50)$$

So we can rewrite problem (49) as follows:

$$\begin{aligned} & \max_{k_{it} \in [0, k_{it}^{limit}]} (1 - s) (1 - \beta) (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\beta}} - a_{it} \\ & \leq \max_{k_{it} \in [0, k_{it}^{limit}]} (1 - \beta) (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\beta}} + R_t (a_{it} - k_{it}) \end{aligned} \quad (51)$$

Now the bank's problem becomes:

$$\max_{k_{it}^{limit}} R_t \left(\int_{i_t \in E} k_{it}^{limit} d_{it} - \int_{i_t} a_{it} d_{it} \right)$$

Subject to:

$$\begin{aligned} & \max_{k_{it} \in [0, k_{it}^{limit}]} (1-s)(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\beta}} - a_{it} \\ & \leq \max_{k_{it} \in [0, k_{it}^{limit}]} (1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\beta}} + R_t(a_{it} - k_{it}) \end{aligned}$$

A.2.2 Entrepreneurs' Borrowing Constraints

After solving the bank's problem, there will be two types of individuals. The first group of people can borrow enough money from the bank to operate the firm at their optimal capital level, so they are unbounded. The other group of individuals' optimal capital is lower than the capital they could borrow, so they are bound by the bank and can only produce at a suboptimal level.

For unbounded individuals, they produce at their optimal capital level as they do in the benchmark model. Their entrepreneurship cutoff is given by equation (36). If the bank wants to prevent them from default, equation (51) can be rewritten as:

$$\begin{aligned} & \max_{k_{it} \in [0, k_{it}^{limit}]} (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\beta}} (1-s)(1-\beta) - a_{it} \leq \\ & (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{* \frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t} \right)^{\frac{\beta}{1-\beta}} (1-\beta) + R_t(a_{it} - k_{it}^*) \end{aligned}$$

$$\Longleftrightarrow$$

$$\begin{aligned} \max_{k_{it} \in [0, k_{it}^{limit}]} (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} (1-s)(1-\beta) \leq \\ (1-\alpha-\beta)(Az_{it})\left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} + (1+R_t)a_{it} \end{aligned}$$

The bank wants to lend as much capital as possible, so the k_{it}^{limit} should be the solution to the following equation:

$$\begin{aligned} (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} (1-s)(1-\beta) = \\ (1-\alpha-\beta)(Az_{it})\left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} + (1+R_t)a_{it} \end{aligned} \quad (52)$$

After solving (52), we have:

$$\begin{aligned} k_{it}^{limit} &= \left(\frac{(1-\alpha-\beta)(Az_{it})\left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} + (1+R_t)b_{it-1}}{(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} (1-s)(1-\beta)} \right)^{\frac{1-\beta}{\alpha}} \\ &= \left(\frac{\frac{1-\alpha-\beta}{\alpha} R_t k_{it}^* + (1+R_t)b_{it-1}}{\frac{(1-s)(1-\beta)}{\alpha} R_t (k_{it}^*)^{\frac{1-\alpha-\beta}{1-\beta}}} \right)^{\frac{1-\beta}{\alpha}} \end{aligned}$$

Since the individuals are unbounded, so we require $k_{it}^* \leq k_{it}^{limit}$. To make $k_{it}^* \leq k_{it}^{limit}$:

$$\frac{s(1-\beta) - \alpha}{\alpha} R_t k_{it}^* + (1+R_t)a_{it} \geq 0 \quad (53)$$

If $s \geq \frac{\alpha}{1-\beta}$, which means financial frictions are small, everyone who passes the entrepreneurial cutoff can get enough capital, and we are back to the benchmark model. So to make sure there exists a separating equilibrium, we also require:

$$s < \frac{\alpha}{1-\beta} \quad (54)$$

What's more, only individuals who satisfy the following condition will be unbounded.

$$\frac{a_{it}}{k_{it}^*} \geq \frac{\alpha - s(1 - \beta)}{\alpha} \frac{R_t}{1 + R_t} \quad (55)$$

Or equivalently, we can rewrite (55) as

$$\frac{a_{it}}{Az_{it}} \geq \frac{\alpha - s(1 - \beta)}{\alpha} \frac{R_t}{1 + R_t} \left(\frac{\alpha}{R_t}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} \quad (56)$$

The intuition behind this is that when the individual has more assets than the entrepreneurial talent, the individual will be unbounded. Next, let's discuss the case when the individual is bound by the bank. In the bound case, we can rewrite the equation (49) as follows:

$$\begin{aligned} (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} (1 - s)(1 - \beta) - a_{it} \leq \\ (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} (1 - \beta) + R_t(a_{it} - k_{it}^{limit}) \end{aligned} \quad (57)$$

The capital limit k_{it}^{limit} is given by the following equation implicitly:

$$s(1 - \beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\beta}} - Rk_{it}^{limit} + (1 + R_t)a_{it} = 0 \quad (58)$$

It's easy to show that k_{it}^{limit} is increasing and concave in both a_{it} and z_{it} .

For the bounded individuals, their entrepreneurial cutoff can be calculated by the following inequality given k_{it}^{limit} :

$$(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{limit \frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (1-\beta) - R_t k_{it}^{limit} - W_t \geq 0 \quad (59)$$

After solving (59), we know that the bound individuals will choose to become entrepreneurs if and only if:

$$Az_{it} \geq \left(\frac{W_t + R_t k_{it}^{limit}}{(1-\beta)(k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}} \right)^{\frac{1-\beta}{1-\alpha-\beta}} \quad (60)$$

A.2.3 Solution Existence & Uniqueness of Entrepreneurial Cutoff for the Bound

If we take the derivative of the right-hand side of (60), we should have:

$$\frac{(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}-1}}{((1-\beta)(k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}})^2} \times ((1-\alpha-\beta) R_t k_{it}^{limit} - \alpha W_t)$$

When $k_{it}^{limit} \geq \frac{\alpha}{1-\alpha-\beta} \frac{W_t}{R_t}$, the right-hand side increases in k_{it}^{limit} . When $k_{it}^{limit} < \frac{\alpha}{1-\alpha-\beta} \frac{W_t}{R_t}$, the right-hand side decreases in k_{it}^{limit} . In the limit, as $z_{it} \rightarrow 0$, k_{it}^{limit} converges to a strictly positive constant, and the right-hand side converges to a strictly positive constant. As $z_{it} \rightarrow \infty$, k_{it}^{limit} again converges to a strictly positive constant, while the right-hand side remains finite. Therefore, the equation has a unique solution.

The law of motion for capital per capita remains the same for the case when there is no financial friction.